You can understand how the metaverse and multiverse compare by looking at the difference between the movies *Avatar* and *Inception*. *Avatar* is set in the metaverse where you immerse yourself in a virtual digital world (Pandora) as an avatar and come back to the universe. *Inception* looks at the multiverse, where you travel as yourself in a digital world (dream sequence), then travel to another world, and so on.

As can be seen in Table 1, the multiverse is a commonly used theme concept for gaming and virtual collaboration platforms, whereas the metaverse has multiple applications in different sectors like virtual assistants, virtual banking, and virtual shopping, to name a few.

The challenges in the metaverse

Closed ecosystem*:* Metaverse is envisioned to connect people. However, its creator Meta is building 'walled gardens' within it, to which only a few tech giants have access. Every company designed for the metaverse environment has its own limitations and boundaries, and is disconnected with other companies. So, making the metaverse interoperable is a challenge.

Virtual reality hardware*:* The only way to access the metaverse is via VR/AR headsets. However, these are too expensive for the average consumer. Also, prolonged usage of these headsets can lead to physical issues such as nausea and eye strains. Headsets developed so far are not very comfortable to wear too.

Addiction*:* The metaverse experience can be very addictive.

Privacy*:* Privacy is a huge concern in the metaverse. VR headsets collect far more information as compared to smartphones or any other smart gadget. For example, companies can get every minute detail about the user such as biometrics, eye movements, and physiological responses. If this information falls in the wrong hands, it can destroy the privacy of the user.

Safety*:* A major issue on the internet is cyber bullying. In the metaverse, the effects of cyber bullying can be even worse as the person is completely immersed in the virtual world. This situation could be catastrophic for kids.

Power consumption*:* Blockchain is an important part of the metaverse. However, blockchains consume a lot of power. As the metaverse grows, it will need more and more power, and that can be a challenge.

Top 10 open source metaverse projects in 2023

Developers are slowly increasing the use of open source to build projects in the metaverse. The top 10 open source metaverse platforms you can explore this year are listed below.

Webaverse*:* This is a great platform for open source learners and metaverse developers. It is an open source web3 metaverse game engine that anyone can host and is fully compatible to run in any web browser. It uses open source tools like three.js and Node.
Website: https://webaverse.com/

GitHub link: https://github.com/webaverse-studios/webaverse

Hypercuber*:* This is a free and open source framework for designing self-hosted metaverse applications. It consists of powerful processing and storage options to design these, and to run AI/VR/AR programs.

GitHub link: https://github.com/hypercube-lab/hypercube

Somnium Space*:* This open source platform looks more like a gaming platform. End users can deploy virtual land, and convert it into powerful residences and commercial establishments.

Website: https://somniumspace.com/

XREngine*:* This open source program for making metaverse development frameworks has powerful features like 3D world-building, real-time chat, user management, etc.

Website: https://kandi.openweaver.com/typescript/XRFoundation/XREngine

Amazon Sumerian*:* This is an open source development platform for making VR/AR/3D apps in any web browser. It makes use of the Babylon.js framework to create 3D scenes. Amazon Sumerian works flawlessly with Amazon services like LEX, Lambda, and the Polly text-to-speech platform.

Website: https://aws.amazon.com/sumerian/

Metaverse	Multiverse
Digital ecosystem	Hypothetical system of multiple digital ecosystems
Assets and human interaction possible	Imaginary world without any assets to own
Can be a mixed reality based universe	Can be a cyclic, simulated, quantum, or holographic universe
Humans are recognised as avatars	Humans take a role play
Commonly used in many sectors like healthcare, banking, and retail for virtual world use cases	Commonly used for gaming and virtual collaboration

Table 1: Metaverse vs multiverse